Lecture Unit 9.3

Sparse matrices and compressed sparse row format

Dense vs. sparse matrices

- In a dense matrix most (or all) of the elements are non-zero
- In a sparse matrix most of the elements are zero
- Sparse matrices often arise in scientific applications e.g. pairwise interactions between resolution elements in a model (finite volume/element methods)
- Storing every element of a sparse matrix would be inefficient so we need an alternative representation

Compressed sparse row format

- "Compressed sparse row format" is one way to efficiently store
- The matrix is stored using three one-dimensional arrays a sparse matrix
- For a matrix with `nrow` rows and `nnz` non-zero elements these arrays are:

  1. The non-zero values: `val(nnz)`
  2. The column indices: `col_ind(nnz)`
  3. Indices in the first two arrays where a row starts:
     `row_ptr(nrow+1)`

see later

Compressed sparse row format

Compressed Sparse Row representation:

$$\begin{pmatrix} 10 & 0 & 0 & 0 & -2 & 0 \\ 3 & 9 & 0 & 0 & 0 & 3 \\ 0 & 7 & 8 & 7 & 0 & 0 \\ 3 & 0 & 8 & 7 & 5 & 0 \\ 0 & 8 & 0 & 9 & 9 & 13 \\ 0 & 4 & 0 & 0 & 2 & -1 \end{pmatrix}$$

Count rows and columns starting from 1:

vals	10	-2	3	9	3	7	8	7	3	8	7	5	8	9	9	13	4	2	-1
col_ind	1	5	1	2	6	2	3	4	1	3	4	5	2	4	5	6	2	5	6

row_ptr	1	3	6	9	13	17	20

By convention the last element in `row_ptr` is set to `nnz+1`
where N_{NZ} = `nnz` = number of non-zero elements

To extract row `i` unpack: `vals(row_ptr(i):row_ptr(i+1)-1)`
`col_ind(row_ptr(i):row_ptr(i+1)-1)`

Compressed sparse row format

Compressed Sparse Row representation:

$$\begin{pmatrix} 10 & 0 & 0 & 0 & -2 & 0 \\ 3 & 9 & 0 & 0 & 0 & 3 \\ 0 & 7 & 8 & 7 & 0 & 0 \\ \boxed{3 \quad 0 \quad 8 \quad 7 \quad 5 \quad 0} \\ 0 & 8 & 0 & 9 & 9 & 13 \\ 0 & 4 & 0 & 0 & 2 & -1 \end{pmatrix}$$

Counting from 1:

vals	10	-2	3	9	3	7	8	7	3	8	7	5	8	9	9	13	4	2	-1
col_ind	1	5	1	2	6	2	3	4	1	3	4	5	2	4	5	6	2	5	6

row_ptr	1	3	6	9	13	17	20

To extract row `i=4` unpack: `vals(row_ptr(4):row_ptr(5)-1)=vals(9:12)`
`col_ind(row_ptr(4):row_ptr(5)-1)=col_ind(9:12)`

Compressed sparse row format

- Can significantly reduce the size of a sparse matrix:

$$\text{Store } 2N_{\mathrm{NZ}} + N_{\mathrm{row}} + 1 \text{ values instead of } N_{\mathrm{row}}N_{\mathrm{col}} \text{ values}$$

- Makes no assumptions about sparsity pattern Doesn’t store any unnecessary elements

- Indirect addressing of `vals` and `col_ind` arrays (slower than addressing directly)

Sparse matrix formats

- Compressed Sparse Row is not the only way to store a sparse matrix
- Other formats can be more efficient if the matrix has a particular structure e.g. non-zero values occur in dense blocks, or bands
- See http://www.netlib.org/linalg/html_templates/node90.html for more information

Unit summary

- Sparse matrices contain mostly zero values
- Compressed Sparse Row format is one way to efficiently store a sparse matrix
- Compressed Sparse Row format stores non-zero values, column indices and row pointers
- Compressed sparse row format is used as a general sparse matrix format in the PETSc library (https://docs.petsc.org/en/latest/manual/mat/)